

flea

NEWS66

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FLEA NEWS is a biannual newsletter about fleas (Siphonaptera). Recipients are urged to check any citations given here before including them in publications. Many of our sources are abstracting journals and current literature sources such as National Agricultural Library (NAL) Agricola, and National Library of Medicine (NLM) Medline, and citations have not necessarily been checked for accuracy or consistent formatting.

Recipients are urged to contribute items of interest to the profession for inclusion herein, including: Flea research citations from journals that are not indexed in Agricola or Medline databases, Announcements and Requests for material, Contact information for a Directory of Siphonapterists (name, mailing address, email address, and areas of interest - Systematics, Ecology, Control, etc.), Abstracts of research planned or in progress, Book reviews, Biography, Hypotheses and Literature Reviews, and Anecdotes. Send to:

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Contents

Announcements
 Meetings
 Featured research
 Directory of Siphonapterists
 Siphonaptera Literature December 2009-June 2010
 Obituaries: Flea larvae experts, R.L.C. Pilgrim and R.E. Elbel

Announcements

On June 30, 2010, Daniel M. de Avelar was scheduled to defend his Ph.D. thesis entitled "Sistemática e análise cladística das species neotropicais do gênero *Tunga* Jarocki, 1838 (Siphonaptera: Tungidae)" [Systematics and cladistic analysis of neotropical species in the genus *Tunga* Jarocki, 1838 (Siphonaptera: Tungidae)] at the Federal University of Minas Gerais, Belo Horizonte, Brasil. Prof. Pedro M. Linardi is his advisor.

CVBD (Canine Vector Borne Diseases) World Forum Meeting was held in New York during April 2010. www.CVBD.org.

The 4th Int. CVBD Symposium was held on March 26, 2009, Seville, Spain.

There was one flea presentation at the meeting:
 Occurrence of *Rickettsia felis* in dog and cat fleas (*Ctenocephalides felis*) from Italy.
 G. Capelli, F. Montarsi, E. Porcellato, G. Maioli, C. Furnari, L. Rinaldi, G. Oliva, D. Otranto.

New book on ectoparasites, including fleas

Yuri S. Balashov. 2009. Acari and Insect Parasitism on Terrestrial Vertebrates. "Nauka" Publishing House, St. Petersburg, Russia. 358 pp., soft cover (in Russian). ISBN 978-5-02-026336-9

The book is reviewed by Igor Uspensky, in *Journal of Medical Entomology* Vol. 47(4): 705-706 (2010).

Laura Fielden (Truman St. Univ.) is collaborating with Boris Krasnov, Ben Gurion University on flea-host specificity.

Meetings

The International Symposium on Ectoparasites of Pets will be held in conjunction with the annual meeting of the American Association of Veterinary Parasitologists and the Livestock Insect Workers Conference July 16-19, 2011. This triad will gather at the Sheraton Westport Hotel in St. Louis, MO, for a joint conference, allowing ectoparasite researchers an opportunity to join with colleagues from the three organizations to share updates and recent developments. Conference information will be available on the AAVP website (<http://www.aavp.org/>). Questions can be directed to Adalberto Perez de Leon (Beto.PerezdeLeon@ARS.USDA.GOV).

Florida Entomological Society 93rd Annual Meeting will be July 25-28, 2010, Jupiter Beach, FL. Wayne.Hunter@ars.usda.gov.

The XIIth International Congress of Parasitology (ICOPA) will be held in Melbourne, Australia, from 15-20th August 2010. Symposia on tick control, control of ectoparasites, vector competence: ticks, mosquitoes, tsetse, tick genomics, understanding the molecular targets of insecticides, mechanisms of resistance in ectoparasites, and vector-parasite interactions, but no symposia on fleas.

PestWorld 2010, October 20-23, 2010, Hawai'i Convention Center.

ESA Annual Meeting, 10001 Derekwood Lane, Suite 100, Lanham, MD 20706, on December 12-15, 2010, in San Diego, California.

Featured Research

Beaucournu, J.-C. & Gonzalez-Acuña, D. 2010. Description de *Trochilopsylla torresmurai* n. gen., n. sp. (Siphonaptera : Ceratophyllidae) du Chili, première puce connue parasite d'oiseau-mouche (Aves : Trochilidae). [Description of *Trochilopsylla torresmurai* n. gen., n. sp. (Siphonaptera: Ceratophyllidae) from Chile, the first flea known to parasitize a hummingbird (Aves: Trochilidae).] Parasite 17, (in press).

Canard, Alain, Jean-Claude Beaucournu, Joël Boustie et Dominique Bernard. 2010. Importance des collections de Sciences naturelles pour la connaissance de la biodiversité. [Importance of natural science collections for the conservation of biodiversity]. *La lettre de l'ocim* No. 129, Mai-Juin.

Bien traitées puis délaissées, les collections de spécimens constituent maintenant une ressource indispensable pour les études sur la biodiversité et ses évolutions : l'exemple de

l'université de Rennes 1 montre que les collections devraient retrouver un statut en rapport avec leur intérêt mais qu'un problème reste à régler, celui de la disparition des chercheurs travaillant dans le secteur de la systématique.

[Well-treated, then forsaken, collections of specimens now constitute an indispensable resource for studies on biodiversity and its evolution: the example of the university of Rennes 1 shows that the collections should find a statute in connection with their interest, but that a problem remains, that of the disappearance of researchers working in systematics.]

Vetter, S. M., J. L. Holmes, J. A. Montenieri, C. B. Graham, M. E. Woods, R. J. Eisen, K. L. Gage. 2010. EFFECTS OF TEMPERATURE ON TRANSMISSION OF *YERSINIA PESTIS* BY THE FLEA, *XENOPSYLLA CHEOPIS*. Program, The 85th Annual Meeting of the American Society of Parasitologists, Colorado Springs, Colorado, June 22 – 25, 2010.

Plague is a zoonotic disease caused by the gram-negative bacterium, *Yersinia pestis*. Infections are transmitted to rodents and other mammalian hosts, including humans, by flea vectors. Traditionally, the mechanism of flea transmission has been assumed to be dependent on overgrowth of the bacteria in the gut of the flea through a process commonly referred to as blockage. This process was previously shown to be temperature-regulated, with blockage failing to occur at temperatures above 30 °C. Recent work, however, has demonstrated efficient flea transmission in the absence of blockage, demanding a re-examination of the transmission capabilities of fleas held at temperatures during which blockage is expected or not expected to form. We infected colony-reared fleas of *Xenopsylla cheopis* with a wild type strain of *Y. pestis* using an artificial feeding system, and maintained these infectious fleas at 10, 23, 27, or 30 °C. Naïve Swiss Webster mice were exposed to infected fleas on days 1-4 post-infection (p.i.), and every 3-4 days thereafter until day 14 p.i. for fleas held at 10 °C, or 28 days p.i. for fleas held at 23-30 °C. Exposed mice were monitored for signs of infection, and transmission was confirmed using *Y. pestis*-specific fluorescent antibody staining or a passive hemagglutination assay/hemagglutination inhibition test on tissues of mice. Fleas maintained at 23, 27, and 30 °C efficiently transmitted infections throughout the study period and median bacterial loads in these fleas remained high (>10⁴ cfu/flea), although bacterial loads began to decline in fleas held at 27 and 30 °C by day 17 p.i. For fleas held at 10 °C, transmission was not observed until day 7 p.i. and their median bacterial loads were significantly higher (>10⁶ cfu/flea) than those observed at the other temperatures. However, the survival of fleas maintained at 10 °C was also lower than fleas held at higher temperatures. Together, our results support the notion that blockage is not necessary for flea transmission at high temperatures, but that temperature does likely influence the dynamics of *Y. pestis* transmission by fleas by perhaps affecting the way the bacterium grows and persists in the flea gut at different temperatures.

Yates, D. L. J.F. Taulman, A.P.G. Dowling, L. A. Durden, and A.F. Newton. 2010. Invertebrates associated with nests of southern flying squirrels – a preliminary report. Program, Missouri Academy of Science, 46th Annual Meeting, Missouri State University, Springfield, MO, April 16 - 17, 2010.

Little has been reported on the arthropod community associated with the nests of flying squirrels. In a study of population dynamics and habitat use of southern flying squirrels under a range of experimental forest management regimes, invertebrates were collected from nests constructed by southern flying squirrels in artificial nest boxes placed on trees in 21 study areas in the Ouachita National Forest of Arkansas. Nests were removed from boxes during the spring of 1993 and 1994. Arthropods were extracted from nests using the Berlese method and were preserved in 95% EtOH. Preliminary evaluation of the invertebrates represented in this collection was made in 1994 and included 29 families from 11 orders of insects, 10 families from 2 orders of arachnids, a diplopod, and a nematode. In 1995 additional analyses revealed representatives of 3 more families of mites and a hemipteran. This paper presents the results of further research carried out during 2009 and 2010 identifying specimens from this invertebrate collection. In addition to numerous fleas and larvae we have identified immature *Ixodes marxi*, a sciurid tick, *Stratiolaelaps* spp., a predatory mite, several midges, a larval Histeridae and a larval *Nemadus horni* (Coleoptera/Leiodidae). Numerous oribatids have been found as well as expected ectoparasitic mite fauna.

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Siphonaptera Literature -December 2009-June 2010

Note: please check all citations for accuracy. Some citations list authors in incorrect order, or were published before, or after, December 2009 to June 2010.

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Obituaries

Editor's note: with the passing of R.L.C. Pilgrim in April, and R.E. Elbel in 2005, the top two leading experts on larvae of fleas are gone. Though there are occasional papers on flea larvae from other researchers, there are no obvious replacements for their broad knowledge. Since *Flea News* did not feature Elbel's obituary previously, the editors thought it would be appropriate in this volume following Pilgrim's.

Robert Louis Cecil Pilgrim (1921-2010)

By T. Galloway

Emeritus Professor Bob Pilgrim died at his home on 7 April 2010, at the age of 88. Bob gained international recognition for his work on flea larvae, especially the work undertaken after he retired from the University of Canterbury in 1983, following a 40-year academic career. At the time of his death, he was completing a monograph on external morphology of flea larvae.

From the time he and his boyhood friends curated their own natural history museum in Christchurch, Bob Pilgrim had a deep love of the natural history of New Zealand, especially the South Island. Even during the early years when his research focussed on muscle physiology, he still worked away on small side projects on the natural history of various local invertebrates. When asked one time why he did not consider working elsewhere in the world, he always maintained that it was the New Zealand fauna where he thought the real work was to be done. Though he carried out independent research projects on such amazing aquatic insects as the ancient scorpionfly, *Choristella philpotti* (Nannochoristidae), or the rarely seen hydrozoan, *Pelagohydra mirabilis*, of local coastlines, and on the insect fauna of Canterbury beaches, his real interest was in parasitic invertebrates. He became knowledgeable about the parasitic copepods of marine fishes. He spent many years working on lice ectoparasitic on birds. In fact, there are at least three species of lice (*Colpocephalum pilgrimi* Price, 1967; *Pseudomenopon pilgrimi* Price, 1974; *Rallicola pilgrim*, Clay, 1972) in New Zealand named after Bob, as well as one species of flea (*Notiopsylla peregrinus* Smit, 1979), one species of feather mite, two species of parasitic copepods (*Mecaderochondria pilgrimi* Ho and Dojiri, 1987; *Schistobrachia pilgrimi* Kabata, 1988) and a cumacean (Crustacea), *Scherocumella pilgrimi* (Jones, 1963).

Bob was the product of the Great Depression. He went to school at Boys' High in Christchurch, but when his father's business failed, he left school to work. Studying part time at Canterbury, he obtained his B.Sc. in 1943. He entered the service in WWII, and was stationed in the army's Force General Hospital in New Caledonia. There he learned a great deal about parasitic infections and tropical diseases. While there, he spent some of his time in U.S. Army medical services tents, always looking to learn more and gain experience. He was later stationed at Ruakura Research Station on the North Island of New Zealand, until after the war, when he assumed a junior lectureship in 1945, now married to his wife, Joy.

Bob and Joy eventually left for England, where Bob spent 1948 to 1951 pursuing studies that would lead to a Ph.D. in nerve/muscle physiology. He returned to the University of Canterbury, where he resumed his responsibilities for teaching in Zoology, and became a full professor in 1969. He was a stimulating and demanding professor, always organized, and always encouraging to the students. One story that persists today is related to his toes. Having worn poorly fitting, uncomfortable shoes for many years during the Depression, Bob's feet suffered permanent damage. His baby toes were extremely painful, and at one point, he had them removed. Never one to waste an opportunity, he kept the toes and decided to have the bones embedded in plastic. He began preparation of the first toe, boiling it in caustic to soften the tissue, when he was distracted, and forgot about the toe. When he finally returned, the liquid had boiled away, and the toe was charcoal in the bottom of the beaker. He was much more careful with the other toe, successfully cleaned it, stained it, and had it embedded, to serve as a paperweight in his office for many years. He used to hand it to people who visited him, and ask them what they thought it was. He was always amused with the occasional physician who was unable to recognize a human toe bone.

When the rigours of administration and teaching began to fragment his time later in his career, he found it increasingly difficult to maintain the strict timing and regimen required to carry out carefully planned experiments in muscle physiology. It was natural for him to pursue

his other interests with increasing intensity, because this work was more flexible in its demands. He could fit this research into the available gaps in his busy schedule. Over time, he came to recognize his niche. The ticks were being amply addressed by Allen Heath in Upper Hutt. Ricardo Palma had come to work with Bob on chewing lice on birds, and when Ricardo was hired at the Museum of New Zealand Te Papa Tongarewa in Wellington, Bob let that area of pursuit go to Ricardo, though for years after, they continued to collaborate. Frans Smit at the British Museum had done the adult fleas found in New Zealand, but the larval stage had hardly been touched. In fact, there were only about half a dozen of the introduced species in New Zealand, such as the cat flea and the rat flea, for which the larvae had been described. Given this enormous gap in our knowledge, it was a logical step to address the description and study of flea larvae.

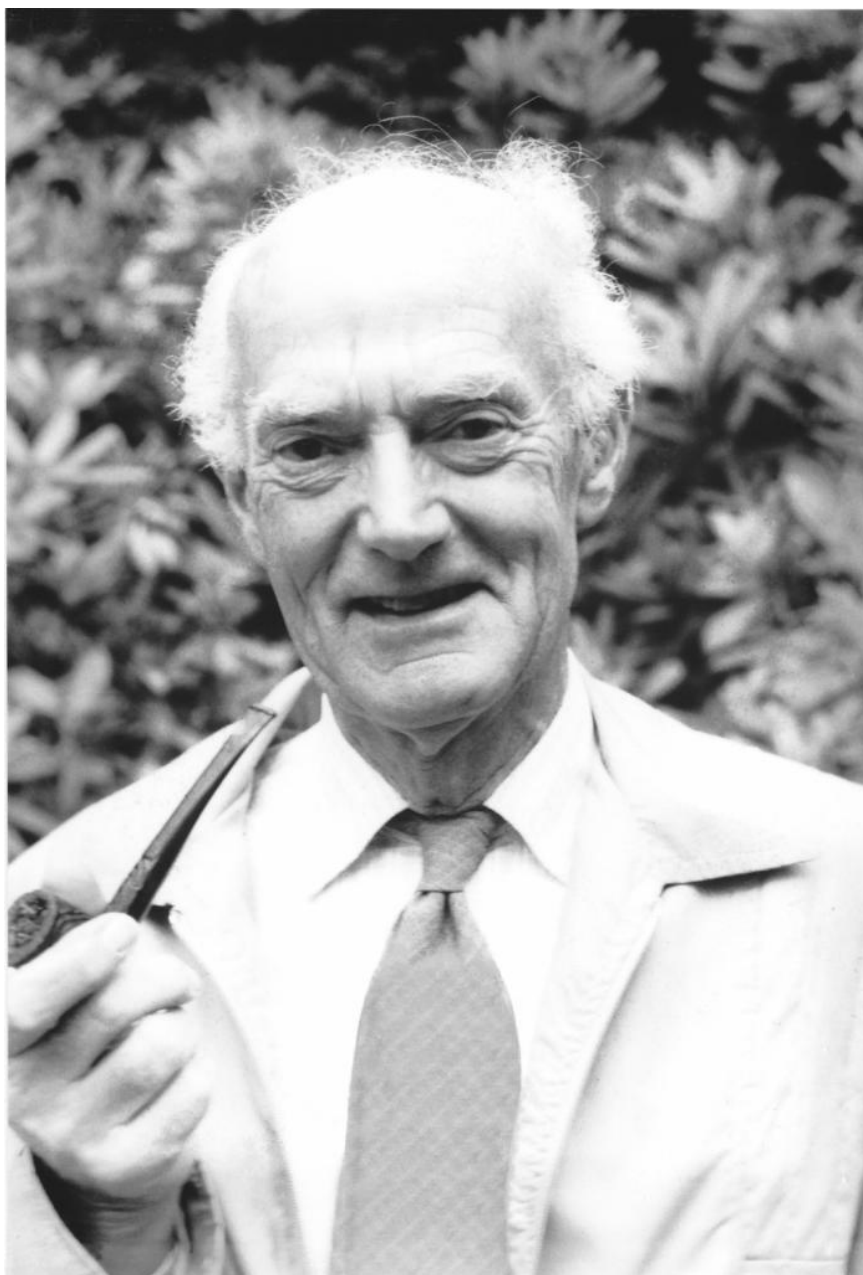


Figure 1. Bob Pilgrim at International Congress of Entomology in Vancouver in 1988.

Bob was a most particular and careful researcher, and as he explored the descriptions of the previously described larvae, he found certain discrepancies with the specimens he had on hand. Using preparation techniques he had learned in studying lice, and applying them in an innovative way to examine flea larvae, he was able to see more easily many critical external characters that had escaped researchers who had gone before. As he accumulated flea larvae for other species, he realised that he would have to retire to be able to commit the time he felt necessary to crack the code. So he retired in 1983 to study flea larvae with the intensity he felt the project warranted. He relentlessly searched the world for more larvae. He struck up correspondence with flea workers, mammalogists and ornithologists alike, appealing to people everywhere to collect flea larvae and to send them to him for study. People mostly responded to his sincere and good-natured insistence, and the flea larvae began to pour in.

As a result, Bob amassed what is almost certainly the largest and most diverse collection of flea larvae in the world. He prepared more than 6,000 slides of larvae and at least another 2,000 of associated adult fleas, representing about 230 species from nearly every flea family known, from every region of the world, including Antarctica. Having seen such diversity, he developed his own system for description of flea larvae, and recognized patterns for identification and classification. From the time he retired, he worked to the end towards a monograph on the external morphology of flea larvae, a tome that will ultimately be the starting point for anyone interested in the study of these curious animals.

Fleas are insects of tremendous medical and veterinary importance, affecting the health and well-being of humans, domestic animals and wildlife. The ability of some species to act as vectors for the bacteria that causes bubonic plague, a disease that has altered the course of human history, makes fleas worthy of study. Bob Pilgrim's research on taxonomy of flea larvae has provided fundamental guidelines of recognition and identification of immature fleas in the absence of adults and provided a solid foundation for future research.



Figure 2. Bob Pilgrim on a field trip to collect fleas from penguin nests at Taylor's Mistake on Bank's Peninsula in New Zealand in 1991.

Publications by R.L.C. Pilgrim

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Robert Edwin Elbel (1925-2005)

By R.L. Bossard

Robert E. Elbel was born in Hannibal, Missouri, on July 8, 1925. His father was a coach at the University of Kansas in track and field, and basketball. Bob, who admitted he was not good enough for the track team, remained interested in fitness and running throughout his life. He collected extensive material on University of Kansas sports.

Elbel's B.A. degree in Entomology, with a minor in Civil Engineering, was completed in 1948 at the University of Kansas, but he realized his greater interest was biology, so he completed a Master's thesis there two years later, with a minor in Botany, on descriptions of several flea larvae and their biology. At that time, a handful of flea species larvae had been described systematically, beginning with Bacot and Ridewood in 1914. He determined that, due to overlap, length was insufficient to distinguish among different instars of flea larvae.

Elbel served in the U.S. Navy during 1943-1945, and worked for the U.S. Public Health Service in Georgia on typhus and dysentery. He applied for the U.S. Agency for International Development program on malaria in Asia and Africa, but heard nothing from them for months, and then a permanent job opened up in the Public Health Service in Georgia. He decided to take it, and told his secretary. She replied, "No, you're not," handing him a newly received letter assigning him to Thailand.

He served two tours in Thailand, the first as a bachelor (1951-1955); the second married (1961-1963), and witnessed a plague epidemic there. Elbel thought the program to eradicate malaria was "a good idea that didn't work." Elbel learned Thai and Laotian. The most useful phrase for him was "kop," which means, "yes," or "okay." When based in Thailand, Elbel traveled on several expeditions to collect ectoparasites from throughout the dense jungles of Southeast Asia and the South Pacific islands. After collecting the hosts and their internal and external parasites, the party would throw the skinned carcasses into a large iron pot with rice, plenty of curry powder, and water over an open fire for a stew that was their dinner each night. He called these experiences "An American in Thailand."

He earned his PhD in Entomology at the University of Oklahoma in 1964 with a dissertation on chewing lice (Mallophaga) of hornbills (Bucerotidae). His major professor was

Cluff E. Hopla (Flea News 63 December 2008, and Journal of Vector Ecology June 2009 34 (2)), who was a new faculty member starting a field survey of Alaskan ectoparasites and tularemia. Elbel married his dissertation typist, Lyda Louise Harris. They had a girl and two boys.

Though Elbel enjoyed living in Thailand, he took a job in 1963 at the U.S. Army Dugway Proving Ground, located in the Great Salt Lake Desert of western Utah, on the eastern edge of the Great Basin Desert. With a team of scientists and technicians, and in conjunction with the University of Utah, Elbel used traps, including stable traps and Center for Disease Control miniature light traps, to collect ectoparasites, their hosts, and microbial pathogens in the Great Basin Desert, especially mosquito arboviruses. Mosquitoes and biting midges were laid on a chill table, identified, counted, and sorted into pooled samples, which were then assayed for viruses using eggs and rabbits. Chill tables are refrigerated, glass-topped tables with dissecting scopes.

He was very interested in rapid identification of arthropods and their hosts ("parataxonomy"). He felt that traditional taxonomic keys required too much specialized knowledge and took too long to be used easily and correctly by technicians. As a result, he published several easy-to-use dichotomous keys using simple diagrams and photographs of species for identifying mosquitoes and mammal hosts.

Elbel helped discover new viruses in Utah. Another finding was an inverse relationship between viral infection in mosquitoes and cyclical rabbit population numbers: when rabbits are abundant, virus incidence in mosquitoes is low. The explanation for this is unclear, but high rabbit or mosquito numbers may dilute the virus, or less likely, the virus is regulating the rabbit population.

The 1970s saw increased environmentalism with Earth Day, the U.S. Endangered Species Act, and U.S. Clean Air and Water Acts, and the emphasis at Dugway turned from disease ecology to environmental impact statements. Elbel found this sort of research uninteresting, so he retired early.

With the help of Dr. Lewis Nielsen, a mosquito taxonomist studying the *Culex* species complexes in the mountain lakes of Utah, Elbel accepted a Research Professor position at the University of Utah Department of Biology in 1974. For the next thirty years, he was paid "a dollar a year," arriving at 9 am and leaving at 6 pm, Monday through Friday, and often Saturdays, and much of the time taking the bus.

The Methodist church was important to him. He often served as usher for the church, and led a church hiking group "The Mountaineers" to Glacier, Yellowstone, and Zion National Parks, among other locations. On backpacking and camping trips, he pursued an interest in insect camouflage and warning coloration using macrophotography, and enjoyed camping and river-running trips on the Snake River with his family.

At the University of Utah, Elbel served as an unofficial "extension entomologist," identifying the innumerable and various insects and spiders that the public brought in, as no one

else in the Department had the time, patience, or expertise. He appeared on television or in the newspaper several times in connection with irruptions of mayfly (Ephemeroptera), Mormon cricket (*Anabrus simplex*), other grasshoppers, and water boatmen (Corixidae). Elbel curated the ectoparasite collections at the Utah Natural History Museum. He had one graduate student, James Kucera, who completed a remarkable study using both light and electron microscopy on the taxonomy and ecology of flea larvae associated with desert woodrat (*Neotoma*) nests. Elbel assisted many others, including the editor (R.L.B.). Elbel frequented the deserts of Utah and Nevada with his colleagues, camping in order to trap hosts and collect ectoparasites.

In 1991, Elbel completed his *magnum opus* with the publication of the Siphonaptera chapter of Immature Insects, edited by F. Stehr (Volume 2, Kendall/Hunt Publishing Company, Dubuque, Iowa). These books were the first to describe the larvae of virtually all the insect orders. Flea larvae are particularly difficult to identify using the light microscope as they have few distinguishing characters, mainly setal lengths and numbers, mandible shape, tooth number, and head capsule shape. In his chapter, Elbel summarized all previous work on flea larvae systematically with a pictorial key for 52 larval species of fleas (of the nearly 300 North American flea species), and descriptions of their ecology and behavior. He corresponded voluminously with Bob Pilgrim.

In this chapter, and two papers following, Elbel speculated that flea larvae:

- Had sufficient morphological features to group them by using light microscopy.
- Did not match current adult systematics.
- Formed distinct subgroups that were not in evidence in adult groups.
- Were selected for in different environments.
- Should have one phylogeny consistent with adult fleas.

In connection with the Bishop Museum in Hawaii, Elbel identified lice (Mallophaga, Anoplura) of the Pacific region. This sort of work is of increasing importance, as many of their hosts are extinct or endangered. During his time in Hawaii, he suffered massive financial losses in his stock investments due to a complex arrangement of “puts and calls” without a “stop-loss.” He blamed deceptive and negligent advice from his broker, and his attempts to get reimbursed and to warn other investors consumed much of his time.

Elbel's research on lice was centered on the “phylogenetic congruence hypothesis” that assumes that host-switching and different rates of speciation between hosts and their parasites are negligible. He advocated using parasites as characters to define species of hosts; that is, when two groups of birds, otherwise indistinguishable, had different species of lice, the birds could be diagnosed as different species because presumably if the two groups could interbreed (as a single species), they would have the same species of lice. Elbel's approach was descriptive “discovery science,” and neither hypothesis-testing nor molecular, which are now popular. Because of these

limitations, he began working with a cell biologist (Ted Gurney) to sequence DNA in order to rapidly identify fleas and mosquitoes, and pathogens they may be carrying, using both preserved and fresh specimens. Elbel and Gurney's procedures remove DNA for molecular identifications without damaging the exoskeleton features required for traditional microscopic identification. Their techniques proved very accurate, revealing not only variation between species but within local populations as useful for population genetics.

On December 1, 2005, Elbel exited his bus at night, crossed the street but not at a crosswalk, and was killed by a hit-and-run driver.

His perfectionism, painstaking attention to detail, and wry sense of humor made him memorable to everyone who knew him. He established the Robert Elbel Scholarship Fund for biology students at the University of Utah (donations may be sent to University of Utah, c/o the Robert Elbel Scholarship, 540 Arapahoe Drive Ste. 250, Salt Lake City, UT 84108). Twenty-six species (*Elbelia*, *elbeli*, *roberti*) are named after Robert Elbel, including fleas, lice, mites, and birds. His publications are listed in Flea News 62 (June 2008).

